

GALAXY SUPERCLUSTERS

22–23 The scale of the universe

24–27 Celestial objects

28–31 Matter

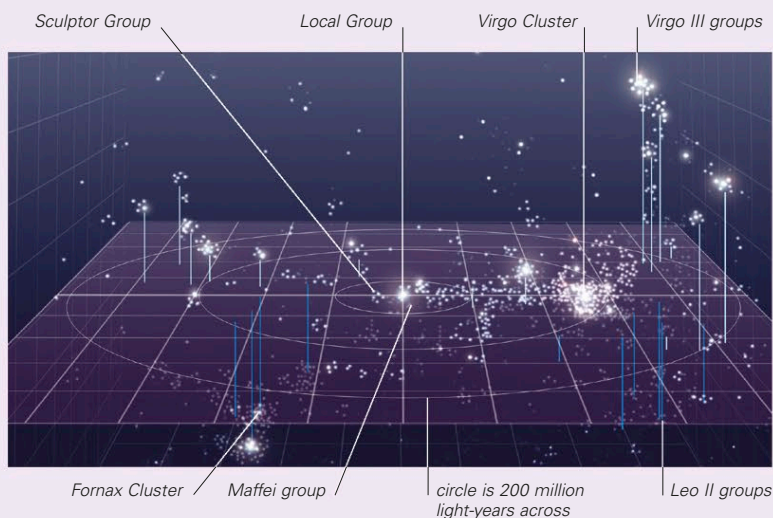
34–37 Radiation

326–327 Galaxy clusters

THE LARGEST-SCALE STRUCTURES in the universe are galaxy superclusters—collections of neighboring galaxy clusters that bunch together in chains and sheets stretching across the cosmos. These structures are echoes of those that formed in the Big Bang, and by studying the universe at these enormous scales, astronomers can learn about the way it formed and our place within it.

GALAXY SUPERCLUSTERS

Just as galaxies are bound together by gravity into clusters, galaxy clusters themselves blur together at their edges to form even larger structures called superclusters. While individual clusters are typically about 10 million light-years in diameter (see p.326), superclusters are typically up to 200 million light-years across and merge with others at their edges. Where superclusters overlap, it is the gravitational behavior of individual clusters that determines to which supercluster they belong. The enormous size of superclusters and the great mass of galaxies in them allows them to modify the cosmological expansion of space (see pp.44–45), resulting in large-scale variations in the movement of galaxies. The best known example of this is a generalized flow of galaxies in our part of the universe, possibly toward a region known as the Great Attractor but more likely toward a more massive supercluster lying directly behind it.



NEARBY CLUSTERS

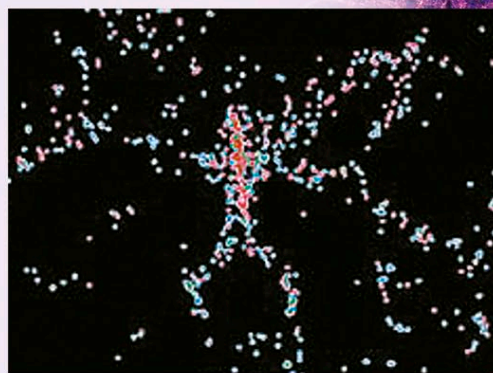
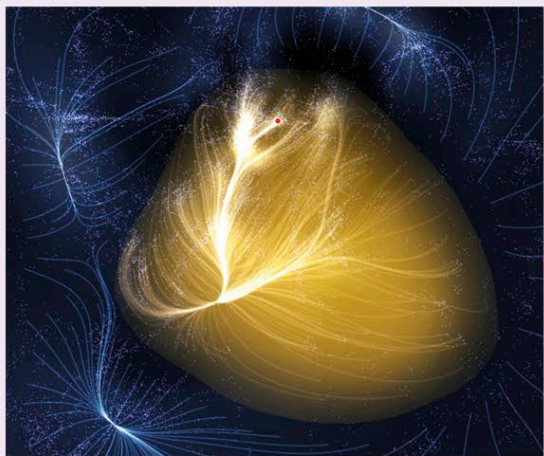
This map shows the distribution of galaxy clusters within 100 million light-years of the Milky Way. Each point denotes a major galaxy—thousands of smaller ones are not included in the map.

THE GREAT ATTRACTOR

The sky in the direction of the Great Attractor is dominated by galaxies in the Norma Cluster, about 220 million light-years away. In 2016, astronomers found evidence for a Vela Supercluster of galaxies lying beyond it.

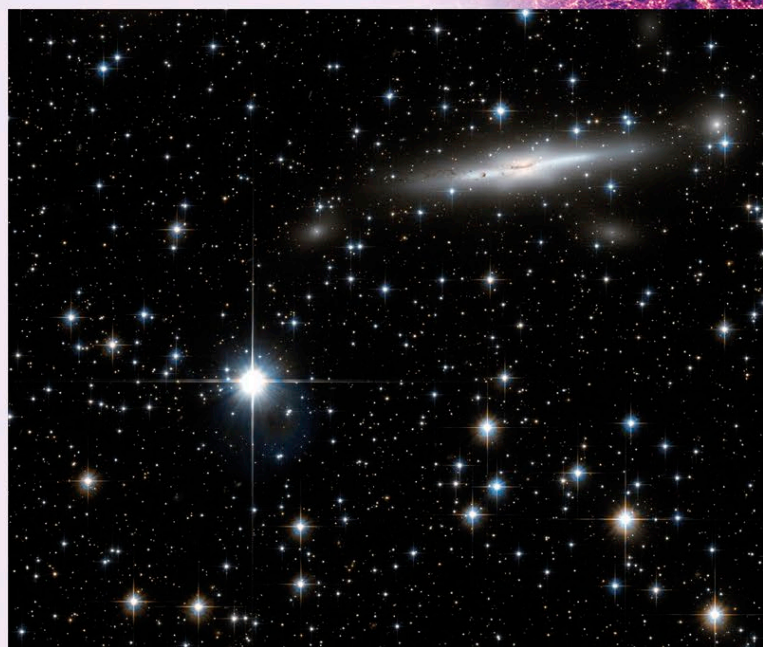
MAPPING LANIAKEA

In this artist's impression of the Laniakea Supercluster, each dot represents one of 100,000 galaxies spread across 520 million light-years of space, with bright areas indicating dense concentrations, such as the local supercluster region. The Milky Way's location is marked by a red dot near the top. Lines indicate the direction in which galaxies are flowing through space, while the yellow region indicates the extent of Laniakea's gravitational dominance.



PLOT OF GALAXIES

This plot of a section of sky out to a distance of 1 billion light-years shows how galaxies cluster on the largest scale.



THE LANIAKEA SUPERCLUSTER

A chain of galaxy clusters links our own small Local Group of galaxies to the Virgo Cluster, some 52 million light-years from Earth. Until recently, this much larger cluster, containing up to 2,000 galaxies, was thought to mark the gravitational heart of a distinct “Virgo Supercluster,” encompassing the Local Group and at least 100 other bright galaxy clusters across 110 million light-years of space. However, the Virgo Supercluster always seemed relatively small and underweight compared to other known superclusters, and a 2014 reassessment of galaxy motions in the local universe saw it subsumed into a much larger supercluster called Laniakea. The Laniakea Supercluster encompasses the former Virgo, Hydra–Centaurus, Pavo–Indus, and Southern superclusters in a volume of space 520 million light-years across and containing 100,000 bright galaxies, with the Great Attractor at its center. The gravitational attraction binding Laniakea's distant extremes is too weak to hold it together against cosmic expansion driven by dark energy, so unlike its component parts, it will ultimately disintegrate.